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A Correction for the α_1 α_2 Doublet in the Measurement of Widths of X-ray Diffraction Lines

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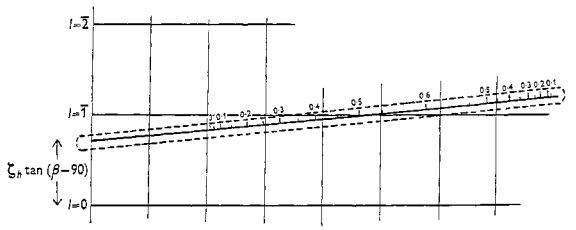
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The accompanying charts give lines of constant D in reciprocal space. They can be used as explained below to find the value of D for any reflexion.

Suppose a normal beam Weissenberg photograph has been taken of the hth layer line of a monoclinic crystal rotated about its a-axis. A straight line is ruled on a strip of transparent material and this is laid horizontally across the chart at the appropriate height, which in this case is $\zeta_h = h\lambda/a$. The points where this straight line is intersected by the curves of constant D are marked on the transparent scale. A drawing of the corresponding zero layer of the reciprocal lattice is now made on a scale 10 cm. = 1, which is the scale of the charts, and a pin passed through the origin of the D-scale so that it pivots about a point distant $\zeta_h \tan(\beta - 90°)$ from the origin, as in Fig. 1. ($\zeta_h \tan(\beta - 90°)$ is the separation in a horizontal plane of the a and a^* axes at height ζ_h.) The value of D for each reciprocal lattice point can then be read off from the scale. The same reciprocal lattice diagram with appropriate D scales can be used to find D for all reflexions which appear on rotating the crystal about this axis.



Example of D-scale

Lines of constant D in reciprocal space

This method has been found to be rapid and convenient in operation, besides being independent of the radius of the camera and the type of photograph (oscillation or Weissenberg), and it avoids the necessity for making separate corrections for the polarization and Lorentz factors. Over most of the region D can be read to $\pm 0{\cdot}01$, the region of reciprocal space where the lines are close together corresponds to back reflexions which are not recorded in any case.

Any other factor which is a function of ξ and ζ can be found in a similar way. One which depends on θ only, such as an atomic scattering factor, is particularly easy to deal with, since the curves of constant f are concentric circles drawn about the origin of the reciprocal lattice.

This technique has been developed in connexion with work on the structures of organic compounds, and I am indebted to the Department of Scientific and Industrial research for a grant which made possible my participation in this research.

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(1) COX, E. G. and SHAW, W. F. B. *Proc. Roy. Soc.* A, **127**, p. 71 (1930).
(2) TUNELL, G. *Amer. Min.* **24**, p. 448 (1939).

Note added in proof. It has recently been pointed out to me that a section of the chart for equi-inclination photographs has been given by Chia-Si Lu (*Rev. Sci. Inst.* **14**, p. 331 (1943)).

A Correction for the $\alpha_1\alpha_2$ Doublet in the Measurement of Widths of X-ray Diffraction Lines

In studies of particle size and lattice distortion effects in crystalline materials by the measurement of X-ray diffraction line widths, an inherent difficulty is due to the $\alpha_1\alpha_2$ doublet structure of the lines. These doublet components overlap when the line becomes broadened and the problem is to obtain the width of the stronger α_1 component in the presence of the α_2.

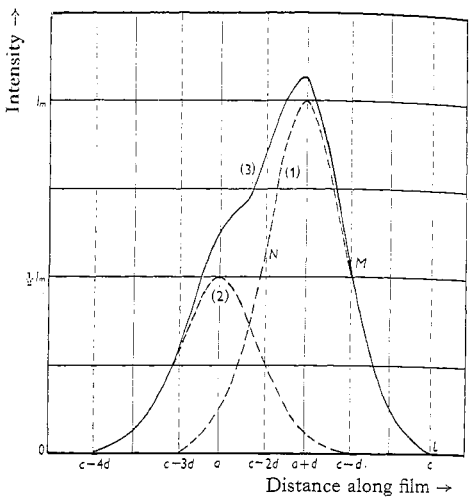
The only data available for the measurement of the width of the single line are:

(*a*) the microphotometer trace of the composite line ($\alpha_1 + \alpha_2$);

(*b*) the separation of the doublet components which can be calculated from the spacing of the reflecting plane, the α_1 and α_2 wave-lengths, and the geometry of the apparatus;

(*c*) the ratio of intensity of α_1 to that of α_2, which has been determined by various workers and may be taken as 2.

The position is illustrated in the accompanying figure in which is shown a plot of intensity (I) against film distance (x) for the α_1 and α_2 components of the diffraction line and their resultant sum $\alpha_1 + \alpha_2$ the latter being the idealized photometer trace of the line.



Relation between intensity and film distance

The α_1 and α_2 components may be represented respectively by the equations:

$$I\alpha_1 = f(x-d), \qquad (1)$$

$$I\alpha_2 = \tfrac{1}{2} f(x), \qquad (2)$$

where d is the doublet separation.

The factor of one-half in equation (2) is a consequence of the fact that

(Intensity of α_1 component)/(Intensity of α_2 component) = 2.

The peak intensity of the α_1 component is

$$I_m = f(a).$$

The integral width B of the α_1 component, which is the quantity that has to be determined, is defined by

$$B = (\text{Area enclosed by } \alpha_1 \text{ curve})/(\text{Peak height of } \alpha_1 \text{ curve})$$

$$= \frac{\int f(x-d)\,dx}{I_m} = \frac{\int f(x)\,dx}{I_m}$$

$$= \tfrac{2}{3}\,(\text{area enclosed by the composite } \alpha_1 + \alpha_2 \text{ curve})/I_m.$$

The measurement of integral width is now reduced to the problem of obtaining I_m.

The most widely used methods for the determination of I_m are those due to Brill(1) in which the profiles of the doublet components are assumed to be triangular and to Jones(2) in which the α_1 and α_2 component curves are assumed to have the form of some known algebraic function.

The first method is obviously unsatisfactory since the lines are not triangular while the method of Jones, in addition to its being tedious, is rather risky since the assumed function is determined from microphotometer records of lines from large strain-free crystals and therefore may not be representative of the shape of the line after broadening. It is the purpose of this note to show that by a simple graphical construction and without any assumptions regarding the shape of the component lines other than those implied in equations (1) and (2), one can not only determine I_m but also resolve the doublet completely.

The procedure is based on the following observation. It is obvious from equations (1) and (2) that if the intensity of the α_1

component becomes effectively zero at $x=c$ then the α_2 curve falls to zero at $x=c-d$. Thus the resultant $\alpha_1+\alpha_2$ curve will be coincident with the α_1 curve LM throughout the range $c-d \leqslant x \leqslant c$. Then from this known part of the α_1 curve one can deduce the remaining shapes of the α_1 and α_2 component curves.

The first step is as follows: Divide the graph into rectangular strips by ordinates whose separation is d, the first ordinate being the line $x=c$. These ordinates are shown dotted in the figure. The α_2 curve in the range $c-2d \leqslant x \leqslant c-d$ can be constructed from the section of the photometer trace included between the lines $x=c-d$ and $x=c$ (i.e. the curve LM) by reduction of the ordinates of LM by the factor of one-half and a translation of this curve through a distance d in the $-x$ direction. Then the α_1 curve in this range (MN) is obtained by subtraction of the α_2 curve from the composite $\alpha_1+\alpha_2$ (i.e. the microphotometer trace).

The procedure is then repeated, the α_2 curve in the range $c-3d \leqslant x \leqslant c-2d$ being obtained by halving the ordinates of the α_1 curve MN and displacing this reduced curve a distance $-d$. The α_1 curve in this range is then obtained by subtraction and the process is again repeated until the curve is completely resolved. This construction of the α_2 curve from that of the α_1 by reduction of the ordinates by one-half, followed by a translation of the reduced curve through a distance $-d$, is easily carried out using a parallel-sided rule of width d (the doublet separation). The right-hand edge of the rule is laid parallel to the I axis intersecting the α_1 curve at a height h above the x axis. The corresponding point of the α_2 curve is then located on the left-hand edge of the rule at a height $\frac{1}{2}h$ above the x axis.

Thus one can, by the above method, determine I_m and hence the integral width of the α_1 line quite simply, or alternatively since complete resolution of the doublet is possible one can determine the half value width of the single line (i.e. its width at one-half of its maximum height) or any other feature of the line which may be required.

The author wishes to thank Prof. J. Neill Greenwood and Dr W. A. Wood for their encouragement and interest in this work.

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(2) JONES, F. W. *Proc. Roy. Soc.* A, **166**, p. 16 (1938).

Added in proof: Since writing the above, the author's attention has been drawn to a paper by Du Mond and Kirkpatrick in *Phys. Rev.* (2) **37**, p. 136 (1931), in which a very similar method has been used in a study of the Compton effect.

NOTES AND NEWS

New Books

Report on Colour Terminology. By a Committee of the Colour Group of the Physical Society. (London: The Physical Society.) Pp. 56. Price 7*s.* net, including postage.

The Committee has dealt with many aspects of the subject, referred to in its terms of reference as 'the recording and co-ordination of terms of accepted significance among colour workers, with recommendations for a consistent terminology'. The problems set by such a study are wide, and it is not surprising that the committee lasted for over 6 years. Its members were for the most part connected with the Colour Group and were able to draw upon a wide range of technical and industrial applications of colour theory and practice. The committee was fortunate in its two successive secretaries, who have been able to impart a unity to the treatment all too rare in reports of such breadth and diversity of interest.

There are few subjects whose terms are of such widespread range in common speech, and yet which call for such precision of definition for particular ends. It is clear that the committee soon found the impossibility of using 'the physicist's system of concepts for defining the terms used in industry' as sole basis of definition. Separate lists for each industry were therefore first drawn up, and it was found possible not only to correlate these individual lists but to build up a correspondence with a broader and more general series of terms used in colour physics. Lists of terms and definitions covering various fields (paints and pigments, dyeing, printing, glass, etc.) drawn up by members or groups specializing in these fields, are set alongside lists of terms used by artists, by physicists and those dealing with colour systems (Munsell and Ostwald) and colour vision, and those used in common speech; all this, together with an analysis of subjective and objective concepts in colour physics, led finally to the system adopted, and to a correlation which, to quote the report, 'was made easier by the surprising degree of agreement which gradually evolved in spite of the original diversities of opinion and—interests served by the individual members'.

This is all well set out in the *Report*, if with varying clarity for different sections, and an admirable summary of the terms used under each heading (Table 1) leads to a final section giving a valuable list of recommended colour terms; this, as well as the whole *Report*, is clarified by a judicious use of different founts, ranging from BOLD CAPS for recommended uses down to *lower-case italics* for those deprecated.

J. H. SHAXBY

The following new books have been received, and unless otherwise stated have been presented by the publishers to the library of the Institute of Physics, where they may be inspected. The Institute assumes no responsibility for the statements given below, information for which has been taken from the books themselves. The amount of space devoted to the notice of any one book must not be taken as a measure of its importance in relation to the others.

Low Temperature Physics, 2nd edition. By L. C. JACKSON, M.Sc. (London: Methuen and Co. Ltd.) Pp. vii + 130. Price 6*s.* net.

The second edition of this little volume, which is one of the publisher's series of 'Monographs on Physical Subjects', follows the plan of the 1st edition, but the various sections have been brought up to date to cover, as far as possible, the large amount of work carried out during the fourteen years since the first edition. The chapter headings are: The Production of Low Temperatures; The Measurement of Low Temperatures; Liquid and Solid Helium; Specific Heats; Electrical Conductivity; Magnetism.

British Sources of Reference and Information. Edited by T. BESTERMAN. (London: Association of Special Libraries and Information Bureaux.) Pp. 56. Price 6*s.* net.

The object of this little guide is to make better known the incomparable wealth of books in British libraries—and above all, the information contained in these books. Hence the booklet opens with a description of the organization through which books in Great Britain can be borrowed, whether from inside or outside the country. This is followed by brief accounts and select lists of the leading library and book organizations, and of the various types of libraries: copyright (national), university, public, special, governmental. A list of the indispensable general works of reference, and a subject index, conclude the guide.